



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 02

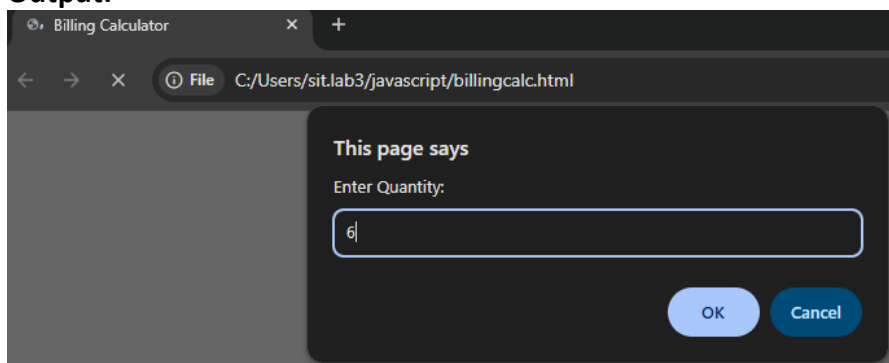
1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring, build a billing calculator with user input.

2. Software/Tools Required: HTML, VS Code.

3. Theory: JavaScript is a scripting language used to make web pages interactive and dynamic. It allows developers to perform calculations, validate user input, display messages, and manipulate webpage content. In this experiment, a **Billing Calculator** is developed using JavaScript. The program accepts user inputs such as customer name, product name, quantity, and price per item through dialog boxes (prompt()). It then calculates the total bill and displays a formatted billing receipt on the webpage. The program demonstrates the following JavaScript features:

- **var** – Used to declare function-scoped variables. It can be redeclared and updated.
- **let** – Used to declare block-scoped variables. It can be updated but not redeclared within the same scope.
- **const** – Used to declare constants whose values cannot be reassigned.
- **Template Literals** – Enclosed within backticks (`), they allow embedding variables directly into strings using `\${}` for easier string formatting.
- **Destructuring** – A feature that extracts values from arrays or objects into separate variables, making the code cleaner and more readable.

4. Output:





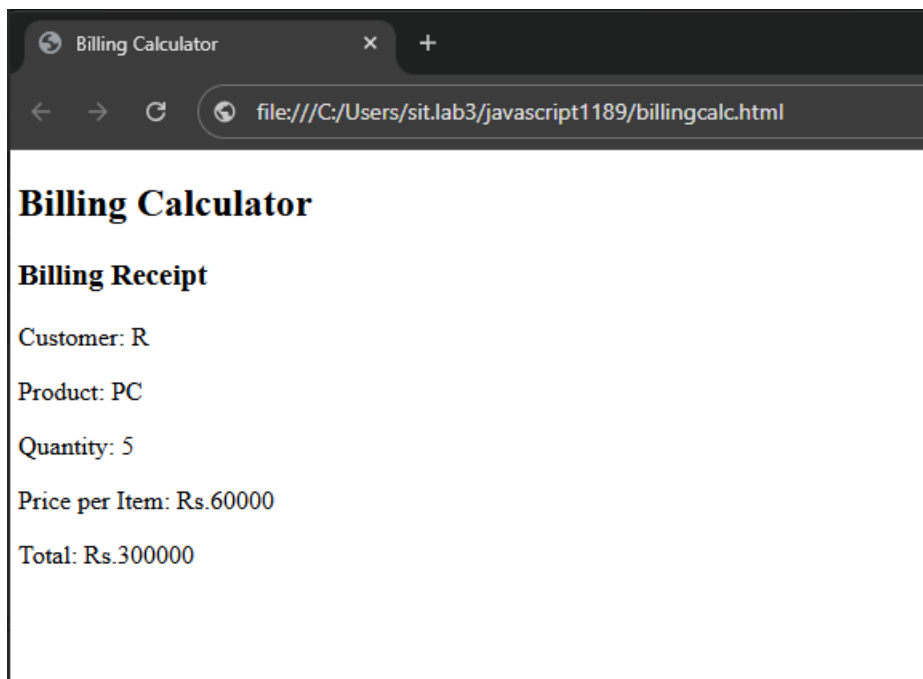
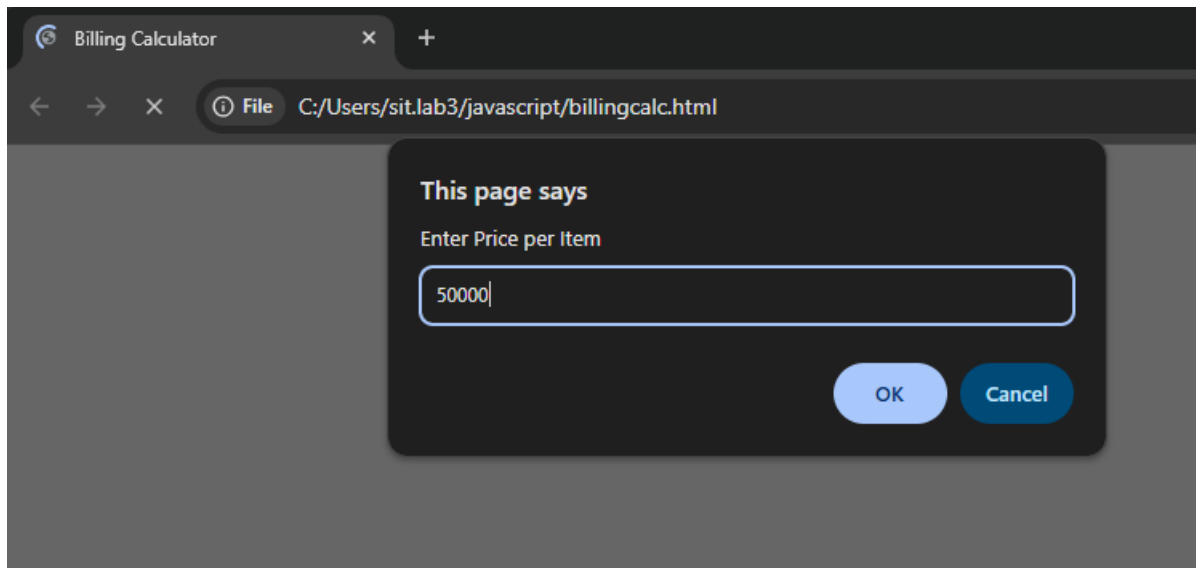
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CASE STUDY 02

Case Study Title: Develop a webpage for an online shopping website using HTML, CSS, and HTML5 tags, and calculate the bill.

Code:

```
Welcome Practical03.html pract04.html department.html
shopping.html > html
2 <html lang="en">
99 <body>
112 <section>
139 </section>
140
141 <footer>
142 © 2026 ABC Online Shopping
143 </footer>
144
145 <script>
146
147 function calculateBill(){
148
149 let laptop=Number(document.getElementById("laptop").value);
150 let headphone=Number(document.getElementById("headphone").value);
151 let mouse=Number(document.getElementById("mouse").value);
152
153 let total=(laptop*50000)+(headphone*2000)+(mouse*700);
154
155 document.getElementById("bill").innerHTML=
156 "<h2>Bill Summary</h2>"+
157 "<p>Laptop: ₹"+(laptop*50000)+"</p>"+
158 "<p>Headphones: ₹"+(headphone*2000)+"</p>"+
159 "<p>Mouse: ₹"+(mouse*700)+"</p>"+
160 "<hr>"+
161 "<h2>Total Bill = ₹"+total+"</h2>";
162
163 }
164
165 </script>
166
167 </body>
168 </html>
```



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Output:

The screenshot displays a web browser window with the title 'Online Shopping Website'. The address bar shows the file path: file:///C:/Users/Hp/Javascript/shopping.html1189Rutva. The website header is dark blue with the text 'ABC Online Shopping' and the tagline 'Shop Smart • Shop Easy'. Below the header is a navigation menu with links: Home, Products, and Contact. The main content area is light gray and contains three product cards. Each card has a title, price, and a quantity input field. The first card is for a 'Laptop' with a price of ₹50000 and a quantity of 1. The second card is for 'Headphones' with a price of ₹2000 and a quantity of 2. The third card is for a 'Mouse' with a price of ₹700 and a quantity of 1. Below the product cards is a green button labeled 'Calculate Bill'. At the bottom of the main content area is a white box titled 'Bill Summary' which lists the items and their prices: Laptop: ₹50000, Headphones: ₹4000, Mouse: ₹700. The total bill is calculated as ₹54700. The footer of the website is dark blue and contains the copyright notice: © 2026 ABC Online Shopping.

Conclusion:

The Online Shopping Website successfully demonstrates the use of HTML, CSS, HTML5 semantic tags, and JavaScript to create a simple and interactive e-commerce webpage. It enables users to select product quantities and automatically calculates the total bill, reducing manual effort and errors. The project highlights the use of webpage structuring, styling, user input handling, event-driven programming, and dynamic bill generation, making it an effective example of a basic online shopping application.